

THE MAXIMUM BINARY CASE OF RIBOT'S ZARISKI-CLOSEDNESS CONJECTURE

ABSTRACT. Ribot conjectured that the orthogonal orbit of the coordinate subspace supported on any partial Latin hyperrectangle is real Zariski closed. We prove the conjecture for every maximum-cardinality support in binary format. Such supports are exactly the two parity classes in $\{0, 1\}^d$. A Fourier change of coordinates diagonalizes the action of $\mathrm{SO}(2)^d$, and character duality for compact tori gives explicit polynomial equations indexed by the integer relations among the resulting characters. In format $2 \times 2 \times 2$, the relation lattice is cyclic and the resulting quartic is a nonzero scalar multiple of Ribot's quartic for the seven-dimensional component. Thus the orbit itself, not merely its Zariski closure, is that hypersurface.

1. STATEMENT

For $x = (x_1, \dots, x_d) \in \{0, 1\}^d$, write

$$e_x = e_{x_1} \otimes \cdots \otimes e_{x_d}.$$

Given $L \subseteq \{0, 1\}^d$, set

$$V_L = \mathrm{span}_{\mathbb{R}}\{e_x : x \in L\}, \quad X_L = \mathrm{O}(2)^d \cdot V_L \subseteq (\mathbb{R}^2)^{\otimes d}.$$

A partial Latin hyperrectangle is a set whose distinct elements have Hamming distance at least two. Ribot conjectured that X_L is Zariski closed for every such support [2, Conjecture 4.4]. Known positive cases include matrices and orthogonally decomposable tensors [1, 2].

For $\delta \in \{0, 1\}$, define the parity support

$$E_\delta = \{x \in \{0, 1\}^d : |x| \equiv \delta \pmod{2}\}.$$

Choose $\Omega \subseteq \{\pm 1\}^d$ containing one vector from each antipodal pair. For a real tensor $T = \sum_x t_x e_x$, put

$$p_\varepsilon(T) = 2^{-d} \sum_{x \in \{0, 1\}^d} t_x i^{|x|} \varepsilon^x, \quad \varepsilon^x = \prod_{k=1}^d \varepsilon_k^{x_k}.$$

Let

$$A : \mathbb{Z}^\Omega \longrightarrow \mathbb{Z}^d, \quad A(m) = \sum_{\varepsilon \in \Omega} m_\varepsilon \varepsilon, \quad \Lambda = \ker A,$$

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and write $m_\varepsilon^+ = \max\{m_\varepsilon, 0\}$ and $m_\varepsilon^- = \max\{-m_\varepsilon, 0\}$. For $m \in \Lambda$, define

$$\Phi_m(T) = \operatorname{Im} \left(\prod_{\varepsilon \in \Omega} p_\varepsilon(T)^{m_\varepsilon^+} \overline{p_\varepsilon(T)^{m_\varepsilon^-}} \right).$$

Each Φ_m is a real homogeneous polynomial in the entries of T .

Theorem 1. *For every $d \geq 2$,*

$$X_{E_0} = X_{E_1} = \{T : \Phi_m(T) = 0 \text{ for every } m \in \Lambda\}.$$

Consequently, X_{E_0} and X_{E_1} are real Zariski closed. Moreover,

$$\operatorname{rank} \Lambda = 2^{d-1} - d.$$

Corollary 2. *Ribot's Conjecture 4.4 holds for every maximum-cardinality partial Latin hyperrectangle in $\{0, 1\}^d$.*

Remark 3. Combining Corollary 2 with Ribot's dimension formula and component classification [2, Lemma 4.5 and Theorem 1.7] shows that X_{E_0} is the unique top-dimensional irreducible component of Ribot's bato variety in $(\mathbb{R}^2)^{\otimes d}$; its dimension is $2^{d-1} + d$.

Remark 4. Although Theorem 1 is stated using all $m \in \Lambda$, finitely many of the equations $\Phi_m = 0$ suffice, since the real polynomial ring in the entries of T is Noetherian.

2. PROOF

Every element of $O(2)$ has the form $R_\theta D^a$, where $R_\theta \in SO(2)$, $a \in \{0, 1\}$, and $D = \operatorname{diag}(1, -1)$. Since D preserves every coordinate subspace V_L ,

$$(1) \quad X_L = SO(2)^d \cdot V_L.$$

Swapping e_0 and e_1 in one factor sends E_0 to E_1 , so $X_{E_0} = X_{E_1}$.

Lemma 5. *For $T \in (\mathbb{R}^2)^{\otimes d}$ and $\varepsilon \in \{\pm 1\}^d$,*

$$p_{-\varepsilon}(T) = \overline{p_\varepsilon(T)}.$$

If $\theta = (\theta_1, \dots, \theta_d)$, then

$$p_\varepsilon((R_{\theta_1}, \dots, R_{\theta_d}) \cdot T) = e^{i\langle \varepsilon, \theta \rangle} p_\varepsilon(T).$$

Finally,

$$T \in V_{E_0} \iff p_\varepsilon(T) \in \mathbb{R} \text{ for every } \varepsilon \in \Omega.$$

Proof. Set $f_\sigma = e_0 - i\sigma e_1$ for $\sigma \in \{\pm 1\}$. Then $R_\theta f_\sigma = e^{i\sigma\theta} f_\sigma$, and the coefficient of $f_{\varepsilon_1} \otimes \dots \otimes f_{\varepsilon_d}$ in T is $p_\varepsilon(T)$. This proves the first two assertions.

If $T \in V_{E_0}$, the defining sum for $p_\varepsilon(T)$ is real. Conversely, if the representative coordinates are real, then $p_{-\varepsilon} = p_\varepsilon$ for every ε . Fourier inversion gives

$$t_x = (-i)^{|x|} \sum_{\varepsilon \in \{\pm 1\}^d} p_\varepsilon(T) \varepsilon^x.$$

Pairing ε with $-\varepsilon$ makes the sum zero when $|x|$ is odd. Hence $T \in V_{E_0}$. $\square$

Lemma 6 (Character duality). *Let $a_1, \dots, a_N \in \mathbb{Z}^d$, define*

$$B : \mathbb{Z}^N \longrightarrow \mathbb{Z}^d, \quad B(m) = \sum_{j=1}^N m_j a_j,$$

and let

$$\phi : \mathbb{T}^d \longrightarrow \mathbb{T}^N, \quad \phi(z)_j = z^{a_j} = \prod_{k=1}^d z_k^{(a_j)_k},$$

where $\mathbb{T} = S^1$. Then

$$\text{im } \phi = \{u \in \mathbb{T}^N : u^m = 1 \text{ for every } m \in \ker B\}, \quad u^m = \prod_{j=1}^N u_j^{m_j}.$$

Proof. The forward inclusion is immediate. For the reverse inclusion, put the integer matrix of B into Smith normal form. The associated unimodular changes of coordinates induce automorphisms of the source and target tori, reducing ϕ to

$$(z_1, \dots, z_d) \longmapsto (z_1^{s_1}, \dots, z_r^{s_r}, 1, \dots, 1).$$

Every power map on S^1 is surjective, so this image is exactly the annihilator of the kernel of the diagonal integer map. $\square$

Proof of Theorem 1. Identify $z = (e^{i\theta_1}, \dots, e^{i\theta_d}) \in \mathbb{T}^d$ with $(R_{\theta_1}, \dots, R_{\theta_d}) \in \text{SO}(2)^d$. By (1) and Lemma 5, a tensor T belongs to X_{E_0} if and only if there are $z \in \mathbb{T}^d$ and $a_\varepsilon \in \mathbb{R}$ such that

$$(2) \quad p_\varepsilon(T) = z^\varepsilon a_\varepsilon \quad (\varepsilon \in \Omega).$$

If (2) holds and $m \in \Lambda$, then the torus factor in the monomial defining Φ_m is $z^{A(m)} = 1$; hence $\Phi_m(T) = 0$.

Conversely, suppose that $\Phi_m(T) = 0$ for every $m \in \Lambda$. Let

$$S = \{\varepsilon \in \Omega : p_\varepsilon(T) \neq 0\}, \quad u_\varepsilon = \frac{p_\varepsilon(T)}{|p_\varepsilon(T)|} \quad (\varepsilon \in S).$$

For every $m \in \ker(A|_{\mathbb{Z}^S})$, extend m by zero outside S . The corresponding nonzero monomial has phase u^m , so its imaginary part vanishes only if $u^m \in \{\pm 1\}$. Thus $q = (u_\varepsilon^2)_{\varepsilon \in S}$ satisfies $q^m = 1$ for every such m . Lemma 6 gives $w \in \mathbb{T}^d$ with $q_\varepsilon = w^\varepsilon$. Choose $z \in \mathbb{T}^d$ coordinatewise so that $z^2 = w$. Then

$$\left(\frac{u_\varepsilon}{z^\varepsilon}\right)^2 = 1,$$

so $p_\varepsilon(T)/z^\varepsilon$ is real for $\varepsilon \in S$ and is zero for $\varepsilon \notin S$. Lemma 5 yields $z^{-1} \cdot T \in V_{E_0}$, hence $T \in X_{E_0}$.

The common zero set of the real polynomials Φ_m is Zariski closed. Finally, $|\Omega| = 2^{d-1}$. The sign vectors span $\mathbb{Q}^d$: if $v = (1, \dots, 1)$ and v_j is obtained from v by flipping its j th coordinate, then the j th standard basis vector is

$(v - v_j)/2$. Choosing one vector from each antipodal pair does not change the span. Therefore A has rank d , and

$$\text{rank } \Lambda = 2^{d-1} - d.$$

□

Proof of Corollary 2. A partial Latin hyperrectangle in $\{0, 1\}^d$ is an independent set in the hypercube graph Q_d . A perfect matching gives $|L| \leq 2^{d-1}$, and the bipartition classes E_0, E_1 attain equality.

Suppose $|L| = 2^{d-1}$ and write $A = L \cap E_0$ and $B = L \cap E_1$. Then $|E_1 \setminus B| = |A|$. Since L is independent, every edge from A ends in $E_1 \setminus B$, so exactly $d|A|$ edges join these two sets. This is the full degree sum $d|E_1 \setminus B|$ of $E_1 \setminus B$; hence there is no edge between $E_0 \setminus A$ and $E_1 \setminus B$. There is also no edge between A and B . Thus $A \cup (E_1 \setminus B)$ has no edge to its complement. Since Q_d is connected, this set is empty or all of Q_d . Accordingly, $L = E_1$ or $L = E_0$, and the result follows from Theorem 1. □

3. THE $2 \times 2 \times 2$ QUARTIC

Under the identification $x_k = i_k - 1$, the even parity class is Ribot's support

$$L_2 = \{(1, 1, 1), (1, 2, 2), (2, 1, 2), (2, 2, 1)\}.$$

Take

$$\Omega = \{\alpha, \beta, \gamma, \delta\}, \quad \alpha = (1, 1, 1), \quad \beta = (1, 1, -1), \quad \gamma = (1, -1, 1), \quad \delta = (-1, 1, 1).$$

The relation lattice is generated by

$$\alpha - \beta - \gamma - \delta = 0.$$

With

$$P = 8p_\alpha, \quad Q = 8p_\beta, \quad R = 8p_\gamma, \quad S = 8p_\delta,$$

Theorem 1 gives

$$(3) \quad X_{L_2} = \{T : H(T) = 0\}, \quad H(T) = \text{Im}(P\overline{Q}\overline{R}\overline{S}).$$

Ribot [2, Example 4.1] gives the Zariski closure associated with L_2 by the quartic

$$(4) \quad F(T) = \sum_{i,j,k \in \{1,2\}} (-1)^{i+j+k} t_{ijk} g(t_{i+1,j,k}, t_{i,j+1,k}, t_{i,j,k+1}, t_{i+1,j+1,k+1}),$$

where shifted indices are read modulo 2 and

$$g(w, x, y, z) = 2wxy + w^2z + x^2z + y^2z - z^3.$$

Proposition 7. *As polynomials in the eight real entries of T , one has $H = 4F$. Consequently,*

$$X_{L_2} = \{T : F(T) = 0\}.$$

Proof. The Fourier formula gives

$$\begin{aligned} P &= t_{111} - t_{122} - t_{212} - t_{221} + i(t_{112} + t_{121} + t_{211} - t_{222}), \\ Q &= t_{111} + t_{122} + t_{212} - t_{221} + i(-t_{112} + t_{121} + t_{211} + t_{222}), \\ R &= t_{111} + t_{122} - t_{212} + t_{221} + i(t_{112} - t_{121} + t_{211} + t_{222}), \\ S &= t_{111} - t_{122} + t_{212} + t_{221} + i(t_{112} + t_{121} - t_{211} + t_{222}). \end{aligned}$$

Substitution into (3) and collection of degree-four monomials gives $H = 4F$; equivalently, the two sides have the same forty nonzero monomial coefficients in $\mathbb{Z}[t_{ijk} : i, j, k \in \{1, 2\}]$. $\square$

Thus Conjecture 4.4 of [2] holds for every maximum-cardinality binary support. The argument does not treat non-parity binary supports or formats with a mode of dimension at least three, and the conjecture remains open in general.

REFERENCES

- [1] A. Boralevi, J. Draisma, E. Horobeț, and E. Robeva, *Orthogonal and unitary tensor decomposition from an algebraic perspective*, Israel J. Math. **222** (2017), no. 1, 223–260.
- [2] Á. Ribot, *Unifying singular value decompositions of tensors via aligned orthogonality*, arXiv:2608.03202v1 [math.AG], 2026.